# RISHABH KUMAR

Senior Data Scientist | AI & Behavioral Simulation | Deep Learning & LLM Calibration

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Senior Data Scientist and Machine Learning Engineer with **7+ years of experience** building production AI/ML systems across identity, behavioral modeling, audience intelligence, and representation learning. Specialized in **foundation representation learning, transformer architectures (cross-attention, mask-aware), synthetic consumer simulation, and empirical calibration** across **350M+ users**. Published author (**ICML 2025 Best Paper**, NeurIPS 2025 Workshop) on modeling user behavior from adaptive surveys. Proven record modernizing legacy systems and shipping scalable deep learning pipelines on AWS SageMaker and Apache Spark.

## TECHNICAL EXPERTISE & COMPETENCIES

**AI & Deep Learning:** Transformers (Cross-Attention, Mask-Aware, Self-Attention), Representation Learning, Self-Supervised Learning, Autoencoders, Deep Matrix Factorization, Multi-Label Classification, Probabilistic Modeling, PU Learning, Anomaly Detection.

**Generative AI & Agents:** Synthetic Audience Simulation, Agent Persona Parameterization, LLM Calibration (Isotonic Regression, Conformal Prediction, Distribution Alignment), LLMOps & Evaluation, In-Context Persona Grounding, LangChain/LangGraph, RAG.

**Languages & Tools:** Python, PyTorch, TensorFlow/Keras, Scala, SQL/PostgreSQL, R, Hugging Face, Spark ML, XGBoost, LightGBM, scikit-learn, Optuna, MLflow, TensorBoard, Docker, Git/GitHub, CI/CD.

**Big Data & Cloud Systems:** Apache Spark, AWS Ecosystem (SageMaker, EMR, Batch, Step Functions, S3, EC2, Lambda, SQS, RDS), Grafana, PII Masking, Consent-Based Consumer Data Pipelines, Privacy-Preserving Identity Resolution.

## PROFESSIONAL EXPERIENCE

### Resonate Networks

Herndon, VA (Hybrid / Boston, MA)

**Senior Data Scientist** `Apr 2024 – Present`

- **Technical Lead for Keystone:** Served as technical lead for Keystone, coordinating a cross-functional team of 6 across data engineering, product, DS/ML, and executive stakeholders to drive technical direction and execution, moving identity intelligence from transient ID-level behavior toward durable person-level representations.
- **Self-Supervised Behavioral Autoencoder (BAE):** Developed a self-supervised transformer/autoencoder foundation model deployed on AWS SageMaker, compressing geo, domain, and NLP-derived category signals into **512-dimensional behavioral embeddings** across millions of daily events for **350M+ users**.
- **Large-Scale Survey Imputation & Persona Reconstruction:** Built a downstream supervised Deep Matrix Factorization model that consumes BAE embeddings to predict **~19K survey attributes (+0.07 mean AUC lift)**, reconstructing sparse adaptive survey responses with population-level calibration across 570K respondents — powering core audience segmentation products and campaign targeting.
- **ICML 2025 Best Paper Architecture (LANTERN):** Formulated and productionized a modular late-fusion cross-attention neural architecture fusing sparse user survey priors with external behavioral features via learned bimodal gating; achieved **+5 point recall lift** and strong performance on rare consumer preferences, directly preserving population heterogeneity.
- **Media Bubble & Ideological Cohort Clustering:** Built *Media Bubble*, an unsupervised behavioral clustering pipeline classifying **~20M devices** into ideological content-consumption cohorts using real-time behavioral event signals, directly informing campaign creative messaging and voter landscape strategy.

**ML Engineer → ML Engineer II → Senior ML Engineer** `Nov 2019 – Mar 2024`

- **Lookalike Audience Discovery (Positive-Unlabeled Learning):** Originated a new lookalike-audience product by adapting an existing pipeline for a new use case, using positive-unlabeled learning to identify high-propensity prospects across **~500M HEM profiles**; outperformed production in **~90% of evaluated audiences** on recall-lift and cumulative-gain metrics.
- **Distributed Spark/Scala NLP Pipeline Modernization:** Migrated a legacy MapReduce NLP pipeline to a distributed Apache Spark/Scala architecture processing millions of daily internet events; reduced compute costs by **~35%** and accelerated processing runtime by **~2.8×**.
- **Large-Scale Production Orchestration:** Streamlined and productionized automated training, inference, and monitoring workflows across AWS SageMaker, EMR, AWS Batch, S3, Step Functions, and SQS for **300M+ active user profiles** across ~1,800 XGBoost models.

### HiLabs Inc.

Germantown, MD

**Data Scientist** `Mar 2019 – Oct 2019`

- Developed risk adjustment and anomaly detection products for healthcare data using frequent-pattern mining on **~3.5B records**, translating high-dimensional data into strategic insights and client-facing recommendations while working directly with executive leadership.

## PERSONAL AI PROJECTS

**HouseRank — Multi-Agent AI Real Estate Decision & Ranking Platform** `Next.js • TypeScript • Multi-Agent Architecture (LangGraph) • OpenAI API • Framer Motion`

- Architected a multi-agent evaluation pipeline coordinating specialized agents (Listing Intelligence, Financial/Investment Modeling, and Lifestyle Fit) with a synthesis evaluator to score properties across **Personal Fit** and **Investment Potential** (PITI+H true affordability & equity growth).
- Built an interactive in-person tour module enabling buyers to log and weight qualitative on-site observations (natural lighting, acoustic/noise levels, layout flow) that trigger dynamic agent re-scoring across aggregate rankings.

**Siyu Chef — Multimodal AI Baby Food Recommendation & Grocery Fulfillment (Android Beta)** `Android • Python FastAPI • Multimodal Vision • SQLite Caching • Amazon PA-API`

- Developed an Android app (in beta) generating stage-appropriate, nutritionally balanced infant meal recommendations personalized to developmental age, nutritional requirements, and **strict allergen/dietary profiles**.
- Built a multimodal AI vision scanner that detects on-hand fridge/pantry ingredients to suggest instant recipes, paired with a FastAPI/SQLite backend powering automated **1-click Amazon Cart grocery fulfillment** via PA-API.

## PUBLICATIONS

**Modeling User Behavior from Adaptive Surveys with Supplemental Context** Best Paper Award

*Aman Shukla, Daniel Patrick Scantlebury, Rishabh Kumar — **ICML 2025 New in ML Workshop** (arXiv:2507.20919)*

Introduced **LANTERN**: a late-fusion cross-attention transformer framework with learned bimodal gating that anchors on sparse survey ground truth while selectively querying external behavioral context to recover long-tail user intent and preserve population heterogeneity.

**Responsible Imputation of User Behavior Surveys via Mask-Aware Transformers** `NeurIPS 2025`

***NeurIPS 2025 Workshop on Reliable ML from Unreliable Data***

Demonstrated robust missing-value reconstruction and uncertainty calibration across non-stationary adaptive questionnaire deployments, maintaining statistical distribution integrity for consumer modeling.

## EDUCATION

**Carnegie Mellon University (CMU)** — Master of Science (M.S.), Machine Learning Specialization — Pittsburgh, PA | Aug 2017 – Dec 2018

**BITS Pilani** — B.E. Chemical Engineering & M.Sc. Biological Sciences (Dual Degree) — Pilani, India | Aug 2011 – Jul 2016

## HONORS & AWARDS

- **Best Paper Award:** ICML 2025 New in ML Workshop (July 2025)
- **Reed Award Winner:** Best Use of Data Analytics / ML in Online Fundraising (Feb 2026)
- **1st Prize Winner:** Resonate Networks Annual Hackathon (June 2021 & July 2025)